(c) Consider a system with the following characteristics:

- L1 cache with 32 cache blocks, block size of 16 bytes. 92% cache hit rate. Cache memory access time = 5ns.
- Main memory access time = 50ns. One Byte is transferred on each memory access.
- Virtual memory system with 1Mbyte virtual address space and 1 Kbyte virtual page size. No TLB.
- Page table resides in main memory, size of one entry = 4 Bytes.
- Access to cache, page table entries and target data in physical memory do not overlap.
- Cache uses Virtual Address for indexing and mapping purpose.
- Assume target data can be found in the cache and/or main memory. Storage memory consideration is not required.

Starting from the virtual address of the target data:

How long does it take to read a byte of data if there is a cache hit?

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Solution:

Cache hit => target data is in the cache→Time needed to read a byte = Cache Access time = 5ns.

PYP

(c) Consider a system with the following characteristics:

- L1 cache with 64 cache blocks, block size of 32 bytes.
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(i) Given that Virtual Page (VP) 1 is mapped to Physical Frame (PF) 9, VP2 to PF11, VP3 to PF13, VP4 to PF15 and VP5 to PF17. VP and FP number is in decimal format. Which virtual address is the data in physical memory address 0x3456 mapped to?

(5 marks)

(ii) Derive the cache mapping format (TAG:BLK:OFFSET). Which cache block is the data in physical memory address 0x3456 mapped to and what is the corresponding TAG value?

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(i) 1 Kbyte page: 10 bits

0x3456 = **0011 01**00 0101 0110 => Physical Frame **13** (1101b).

=> Virtual Page **3** (VP3 to PF13) **(2marks)**

=> Virtual Address (1Mbyte: 20 bits) **0000 0000 11**00 0101 0110 or 0x00C56 **(3marks)**

(ii) 32 bytes Offset => 5 bits offset

64. Cache blocks => 6bits block field

Tag bits = 20-6-5 = 9bits

=> Cache Mapping Format = **9:6:5** **(2marks)**

Using Virtual address for cache mapping

=> **0000 0000 1** **100 010** 1 0110

=> **Cache block = 34**, **TAG = 1** **(3marks)**